

Ultimate AI-Powered Trade System - User Manual

Ultimate AI Trade System - Extended Manual

System Overview

This is an automated trading system for Nifty options using AI, price action, real-time news sentiment, and Upstox API. It performs signal detection, model prediction, risk evaluation, trade execution, and daily reporting.

System Architecture

Core Modules:

- Trade Engine (technical + AI logic)
- ML Model (XGBoost classifier)
- Broker API (Upstox)
- Dashboard (Streamlit)
- Risk Monitoring & Logging
- Real-time Sentiment Analysis

Data Sources:

- Yahoo Finance (market data)
- NewsAPI (sentiment headlines)
- Upstox (execution + instruments)

Setup Instructions

1. Install dependencies:

```
pip install -r requirements.txt
```

2. Configure:

- Upstox token
- Email (SMTP)

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- Streamlit password

3. Train model:

```
python ai_trade_predictor_xgb.py
```

4. Run bot:

```
python ultimate_trade_bot.py
```

5. Run dashboard:

```
streamlit run nifty_streamlit_dashboard.py
```

News Sentiment Integration

File: news_sentiment_module.py

- Uses NewsAPI to fetch latest headlines
- TextBlob for NLP polarity scoring
- Returns a float score between -1 (very negative) and +1 (very positive)

Uses:

- Add to trade decision filters
- Annotate daily report
- Monitor news-based risk

Risk Monitoring

Tracks:

- Max drawdown
- Win/loss streaks
- Average RR

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- Model performance (F1, Accuracy)
- Market sentiment log

Daily Automation

- Runs each morning pre-market
- Generates:
 - Trade log Excel
 - PDF summary
 - Chart snapshot
 - Option call/put strategy
 - Email delivery

Auto-learns via weekly model retrain

Cloud & CI/CD

- GitHub Actions: `.github/workflows/ci.yml`
- Docker: `Dockerfile`
- Streamlit Cloud: app-ready with `secrets.toml`

Deploy securely with password-gated UI

Customization

You can extend this system by:

- Adding more indicators (SuperTrend, ADX)
- Integrating with Telegram bot
- Using Zerodha Kite API
- Adding sentiment history chart

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Final Notes

The system is modular, testable, and scalable.

With AI filtering, news sentiment, and automated execution,
it delivers edge and consistency for intraday or positional Nifty trades.

Use responsibly with capital at risk.